

Temporal-Support Identifiability in Event-Detection Evaluation Pipelines

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Abstract

001 *An event-detection benchmark identifies each scored sam-*
002 *ple by one nominal timestamp, but a recurrent detector*
003 *forms its prediction from a longer history, and the bench-*
004 *mark does not record how long. We state the missing re-*
005 *quirement as a property of the protocol, temporal-support*
006 *identifiability: the benchmark variables must uniquely de-*
007 *termine the temporal evidence supporting a reported pre-*
008 *diction. This makes effective temporal support part of the*
009 *benchmark specification rather than a hidden property of*
010 *one evaluation implementation. A released SSM-ViT eval-*
011 *uation pipeline provides a direct counterexample. At the*
012 *same evaluated timestamp, with frames, weights, labels and*
013 *evaluator fixed, moving only the chunk boundary changes*
014 *the recurrent history from one to twenty-one windows within*
015 *the pooled score. The same move changes S5-B by 4.20*
016 *mAP over all labeled boxes and by 5.23 ± 0.49 mAP on*
017 *the velocity-evaluable subset. Because identifiability is a*
018 *uniqueness property, one controlled pair in which identical*
019 *recorded protocol variables yield different effective support*
020 *and different scores is sufficient to show that the protocol*
021 *under-specifies the score. Two supporting diagnostics re-*
022 *ported in the supplement, an input-sensitivity measurement*
023 *on the RVT predictor and an exposure analysis of DSEC*
024 *label timing, bound the interpretation without being used*
025 *as recurrent-support estimators. Over all labeled boxes the*
026 *validation-split ordering of the 5 released checkpoints is it-*
027 *self reproduced in only 42.5 % of sequence-bootstrap repli-*
028 *cates, so that ordering cannot certify a timing convention as*
029 *immaterial. Timing comparisons between recurrent detec-*
030 *tors therefore require the support that produced each score*
031 *to be stated and equalized, a condition an event-detection*
032 *protocol can record with its labels.*

033 1. Introduction

034 Two recurrent detectors evaluated at the same nominal
035 timestamp are not comparable in time unless the history

each one actually used is also identified.

An event-detection benchmark defines a sample by a nominal timestamp. A prediction $\hat{s}$ is scored against the ground truth s^* carried at query time t through a function $\rho(\hat{s}, s^*)$ of position, shape and class. Beyond selecting the label at t , that function contains no variable for the temporal evidence behind either argument, so any temporal discrepancy enters the score disguised as displacement.

A recurrent detector does not read only the instant the benchmark names. It computes $\hat{s}$ from an input window and from state carried across earlier windows. The label it is compared against comes from an exposure of finite width, an annotator or tracker working on the exposed image, and a geometric transfer into another sensor's frame. The evidence on both sides is an interval, while the protocol records an instant.

Three terms carry this throughout. The *nominal timestamp* is the instant at which the protocol pairs a prediction with a label. The *effective temporal support* is the past that can influence the prediction, including state carried by recurrence. The *label-acquisition support* is the interval over which the label's evidence was acquired. A nominal timestamp determines neither of the other two uniquely, and a temporal descriptor may be reconstructable from released artifacts yet remain unreported or heterogeneous within one pooled score. We call this protocol property temporal-support identifiability: whether the released protocol and artifacts uniquely determine the temporal quantities a reported score represents.

We test it with a paired intervention on the released SSM-ViT evaluation pipeline. In that implementation the nominally carried state is inert across evaluation chunks. The recurrent history available to a prediction is a deterministic function of chunk position, and ranges from one to twenty-one windows inside one pooled score. Moving every chunk boundary 5 windows later re-scores the same frames under a longer history at the same evaluated timestamp, with frames, weights, labels and evaluator fixed (3 574 paired boxes, cluster bootstrap over 406 sequences).

Effective temporal support alone changes S5-B by 4.20

mAP over all labeled boxes and 5.23 ± 0.49 mAP on the velocity-evaluable subset (Sec. 3). A second checkpoint of the same released pipeline moves by 4.10 and 4.81 points. Since temporal-support identifiability is a uniqueness property, this controlled violation is sufficient to show that the nominal timestamp does not uniquely specify the temporal evidence represented by that score. The consequence is at the level of the protocol rather than of one implementation: where the protocol does not record the support behind a score, the reported score is underdetermined. We measure that condition on the two protocols examined here and do not assert it of event-detection scores in general.

Contributions.

- *Temporal-support identifiability*: an evaluation property separating the nominal timestamp a protocol records from the effective temporal support and label-acquisition support a reported score represents.
- A paired intervention that isolates recurrent support: the frames, weights, labels, evaluator, and evaluated timestamp are fixed, while only the recurrent history changes.
- Its measured consequence: support the protocol does not record moves a pooled leaderboard-style score by 4.20 mAP overall and 5.23 mAP on the velocity-evaluable subset. The same benchmark reproduces its validation-split ordering over all labeled boxes in only 42.5 % of sequence-bootstrap replicates. Predictor-side and label-side audits delimit the result.

Work on event-detection timing takes that index as given and improves what is measured against it. Sensor latency is calibrated away [25, 29], compute latency is folded into average precision [9, 30, 35], and the temporal window is redesigned [24, 26]. Each changes what the predictor does at a fixed benchmark index, whereas the object measured here is that index itself and the support a score reported under it pools. The two datasets examined here instantiate this protocol condition. How often it holds elsewhere is not measured.

Supporting audits. Two audits, summarized in Sec. 4 and reported in Supplement Secs. 15 and 16, delimit this result and are not used as recurrent-support estimators. Ablating each input bin of the event branch examined here [24] on real validation samples puts the ablation-sensitivity centroid at -23.81 ms, and the regression-based temporal coefficient estimated on nominal-time-matched detections is -2.40 ± 7.18 ms and is compatible with zero. Both describe predictor-side timing, and the nominal timestamp encodes neither. On the label side, across the eighteen DSEC training sequences whose exposure files resolve, published exposure width varies by a factor of 127 against a frame period fixed at 50 ms, and within-exposure event timing carries an illumination-locked component, so it is not at-

tributable to object timing without separating that component first.

2. Related work

Sensor and runtime latency. Prior work calibrates sensor latency [25, 29], incorporates runtime latency into average precision [9, 30, 35], or proposes temporal tolerance [36]. These methods address the time from an event to an output and take the benchmark’s temporal index as given. We instead measure that index, testing whether a released benchmark specification uniquely determines the temporal quantities behind its score.

Temporal windows and benchmark identifiability.

RVT and SSM-ViT redesign or parameterize temporal windows [24, 26], while adaptive-window methods compare alternative partitions [28]. Event- RGB frequency mismatch and exposure duration have also been noted [22, 23, 37]. Both released implementations leave a frame’s position inside the evaluation chunk unspecified by the protocol rather than fixed by the model. The recurrent state entering a chunk is the state saved at the end of the previous chunk, and is reset only at the first sample of a sequence. Under streaming sampling every timestep emits predictions, while otherwise only the last does (RVT modules/detection.py:232,243 and SSM-ViT :315,318). LEOD [27] states that event detectors take only events triggered before the query time and builds a time-flip augmentation on that premise. It does not measure where inside that window the evidence sits. This is a reading of the released code and text, not a measurement of those systems. We read the released per-frame windows and exposures and separate predictor support, output-time interpretation, and label-acquisition support.

Benchmark reliability and evaluation analysis.

Prior work studies detector calibration, stratified model evaluation, label-induced benchmark instability, and event-vision evaluation protocols [31–34]. These analyses examine uncertainty or error in reported performance, whereas our focus is the temporal support and acquisition semantics that a nominal detection timestamp leaves implicit.

Label time and attribution.

Waymo and annotation studies estimate per-object capture or annotation offsets [5–7]. Event datasets expose per-box time fields, but their degrees of freedom are not necessarily specified. LET-3D-AP and TIDE/DETAD/HOTA decompose or bucket spatial error while taking the label clock as given [1, 11–14].

Event benchmarks and exposure.

Gen1, 1 Mpx, DSEC-Det and Ev-3DOD use asynchronous measurements, rec-

174 tified frames, or interpolated labels [16, 19–21]. LEOD
175 establishes causal input windows [27]. The present work
176 reads the released window and exposure artifacts as proto-
177 col evidence and tests whether the resulting temporal quan-
178 tities are uniquely specified.

179 3. Primary evidence: the same-frame chunk- 180 boundary intervention

181 The state the SSM-ViT release carries between evaluation
182 chunks does not affect any of its outputs. It enters as a multi-
183 plicative factor on the first coefficient of an associative scan
184 on which the outputs do not depend. Under the released
185 streaming evaluation this bounds the recurrent history avail-
186 able to a detection to the current 21-window chunk. That
187 history ranges from one window to 21 with the detection’s
188 position in the chunk, and the reported number averages
189 over all 21 positions. Chunk position is a deterministic
190 function of the release’s own index files and is recorded
191 nowhere in the evaluation protocol.

192 **The same-frame intervention.** The intervention moves
193 every chunk boundary 5 windows later and changes nothing
194 else: the same frames, the same weights, the same labels,
195 the same evaluator, and the evaluated timestamp of each
196 frame unchanged. 3 574 labeled boxes are scored twice
197 under this shift, once with one to four windows of recur-
198 rent history and once with seventeen to twenty. They are
199 compared paired within frame with a cluster bootstrap over
200 the 406 sequences. The long-history arm gains 5.23 ± 0.49
201 mAP points for S5-B and 4.81 ± 0.52 for S5-S on the
202 velocity-evaluable subset, and over every labeled box the
203 S5-B gain is 4.20 ± 0.36 points. Mean detection confidence,
204 which uses no labels, moves with it by 0.0579 ± 0.0048 and
205 0.0509 ± 0.0035 . Fig. 1 (bottom) draws one such pair. No
206 quantity the protocol records distinguishes the two arms.

207 **Graded response and what moves.** The size of the
208 boundary shift is a free parameter. Re-scoring the same
209 3 574 frames after shifting the boundary so that the starved
210 block gains +6, +11 and +16 windows of history changes
211 the velocity-subset mAP of S5-B by 3.73 ± 0.35 , 4.91 ± 0.50
212 and 5.23 ± 0.49 points, and that of S5-S by 3.54 ± 0.39 ,
213 4.29 ± 0.52 and 4.81 ± 0.52 . The zero-shift control, which
214 re-runs the release’s own boundary placement, returns 0.00.
215 Confidence on detections matched to a label rises by 0.0409
216 and 0.0519, and total output falls by 3.49 and 5.09 boxes per
217 frame, so both terms of precision move. The curve is mea-
218 sured at four amounts on one dataset and one architecture
219 family, and a saturation point is not estimated (Supplement
220 Sec. 13).

The observational contrast across chunk positions. Re-
constructing chunk position over 20 296 labeled frames of
the released evaluation and scoring every labeled box, S5-B
reaches 34.445 mAP on the frames given one to four win-
dows and 40.226 on those given seventeen to 21 (Fig. 1,
top). This contrast compares different frames, so it is dif-
ferenced against the three RVT checkpoints, evaluated on
the same frames with a state that crosses chunk bound-
aries. That controls for frame-position effects shared with
them. It leaves 5.32 ± 0.53 points for S5-B ($z = 10.0$)
and 4.37 ± 0.61 for S5-S ($z = 7.2$), against placebo differ-
ences among the RVT checkpoints of at most 0.67 points
($z = 1.56$). Permuting chunk position within each sequence
puts both SSM values at least 7.6 standard deviations out-
side their own null and every placebo inside it. The two
readings agree in sign and magnitude, and the paired inter-
vention carries the claim because it holds the frames fixed
(Supplement Sec. 13).

The control, measured rather than assumed. That dif-
ferencing treats the RVT checkpoints as a control because
their state survives a chunk boundary. The same-frame in-
tervention applied to RVT measures that property. Driven
on the SSM chunk grid, RVT reproduces its stored per-
window detections bit for bit at zero shift, and the same
3 574 frames are re-scored with every boundary moved 5
windows later. With the released state carried across the
boundary the paired change is -0.01 ± 0.03 points for RVT-
s ($z = -0.3$) and 0.05 ± 0.04 for RVT-b ($z = 1.3$). Drop-
ping the state at every chunk start, the condition the SSM
release is already in, moves the same weights on the same
frames by 5.39 ± 0.45 ($z = 12.0$) and 6.91 ± 0.64 points
($z = 10.9$). Random-frame nulls are centered within 0.112
points of zero. The boundary effect follows whether the re-
current state survives the boundary, not the architecture, and
the control the difference-in-differences relies on holds as a
measurement (Supplement Sec. 13).

What the pooled score reports. One released number
therefore pools detections whose recurrent history differs by
a factor of 21. Restricting evaluation to the frames receiving
17–21 windows yields an S5-B score 1.75 points above the
pooled released-evaluation score. Two systems compared
under this protocol at the same nominal timestamps need
not have been given the same temporal evidence, and the
protocol records no variable by which a reader could tell.

265 4. Supporting audits

Two audits delimit how the intervention of Sec. 3 is read.
Each measures a timing descriptor that the nominal times-
tamp does not encode, and neither estimates the recurrent
support that the intervention manipulates. Their derivations,

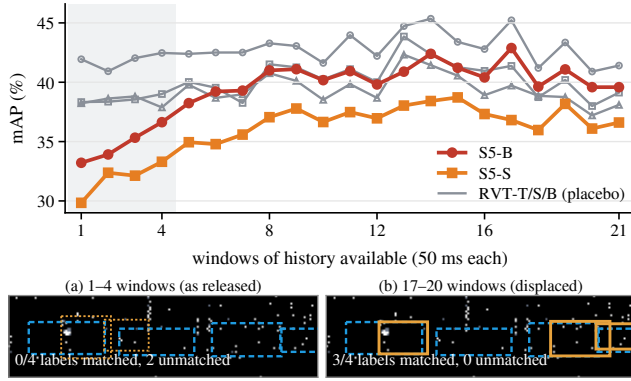


Figure 1. **Top:** benchmark score against the recurrent history the released streaming evaluation made available, at each of the 21 chunk positions, over every labeled Gen1 box with no displacement applied. Shaded: the one-to-four-window block the release scores together with the rest. Between that block and frames given seventeen or more windows, the two SSM checkpoints rise by 5.78 and 4.83 points; the three RVT checkpoints, scored on the same frames with state that crosses chunk boundaries, rise by at most 0.80. **Bottom:** one frame of the same-frame intervention on S5-B. Changed: the chunk boundary, and with it the recurrent history behind the detection. Fixed: frame, weights, labels (dashed), evaluator and evaluated timestamp. Solid: a detection matched to a label of its class at $\text{IoU} \geq 0.5$; dotted: unmatched; drawn above confidence 0.3 and scored at every confidence. The example is illustrative; over all 3574 frames the same intervention changes S5-B by 4.20 mAP over all labeled boxes and 5.23 ± 0.49 mAP on the velocity-evaluable subset (Supplement Sec. 13).

figures and tables are in Supplement Secs. 15 and 16.

Latent quantities. A benchmark releases one tuple per evaluated instant. The tuple holds a nominal timestamp, the labels asserted at it, the predictions scored against them, and the score. Three temporal quantities are latent in that tuple and are not recorded in it: the temporal support of the predictor’s input and carried state, the effective state time its output refers to, and the acquisition support of the labels. Call such a quantity *protocol-identifiable* when the released protocol and artifacts determine it uniquely. Under this definition, input and exposure intervals are protocol-identifiable, whereas effective predictor timing, recurrent context represented by a pooled score, and within-exposure attribution are not.

Predictor side. The released RVT configuration [24] indexes each ten-bin, 50 ms input window by its end, and places that end on a label time. The window’s uniform-weight centroid therefore lies 25 ms before the instant at which the label asserts a state. Ablating each bin of the released `rvt-t` checkpoint over 480 Gen1 validation samples, with the incoming recurrent state held fixed, puts the ablation-sensitivity centroid at -23.81 ms. A regression of detection-center residuals on label velocity, which carries the temporal calibration of [8] to a box center and is fitted on nominal-time-matched detections, gives a coeffi-

cient of -2.40 ± 7.18 ms. The two are distinct operational summaries. Their point estimates are 26.2 ms apart, and the difference is not an estimate of architectural latency. Over a 1000 ms horizon the newest window accounts for 22.7 % of the summed window-ablation sensitivity, so the newest-window centroid does not describe the full recurrent support. Displacing ground-truth centers across the measured label-to-centroid interval changes mAP on the velocity-evaluable subset by 0.04 points and does not reorder the 5 released checkpoints, an effect at the benchmark’s sampling-resolution floor. That near invariance follows from Gen1’s speed composition. It characterizes the metric’s resolution and does not show the absence of a temporal discrepancy (Supplement Secs. 2, 11–12, 14, 15).

Label side. Across the eighteen DSEC training sequences whose exposure files resolve [15], published exposure width varies by a factor of 127 against a frame period fixed at 50 ms, and 32.5 % of frames in six sequences are pinned at an auto-exposure ceiling of $14996 \mu\text{s}$. In those sequences the event-rate spectrum carries the 100-Hz line and its harmonics, which the daytime control lacks. Within-box dispersion of event time inside one exposure is $206.6 \mu\text{s}$ against an analytic null of $93.0 \mu\text{s}$. A frame-level permutation null gives $78.6 \mu\text{s}$, and the excess remains after the clean-pixel control. Random-position boxes show that the dispersion is not specific to annotated object regions. Within-exposure event timing therefore contains an illumination-locked component, and it cannot be attributed to object timing without separating that component first (Supplement Secs. 9 and 16). DSEC-Det ships more than one annotation version [18], and Supplement Sec. 8 names the one measured here.

5. Scope and limitations

Each supporting audit (Sec. 4, Supplement Secs. 15 and 16) is conditional on its stated instrument: local window-ablation sensitivity, nominal-time association under the reported regression specification, and a local constant-velocity displacement surrogate. The primary chunk-boundary intervention is established on two released SSM-ViT checkpoints under their released streaming evaluator, the second testing that the violation is not confined to one weight set. Gen1 and DSEC are separate timing audits, not premises of that counterexample.

6. Conclusion

At the same evaluated timestamp, with frames, weights, labels and evaluator fixed, moving only the chunk boundary shifts S5-B by 4.20 mAP over all labeled validation boxes and 5.23 mAP on the velocity-evaluable subset. One pooled score in the released SSM-ViT pipeline combines predic-

344 tions carrying one to twenty-one windows of recurrent his-
345 tory. The chunk boundary is an unreported degree of free-
346 dom of the protocol. It changes the recurrent support behind
347 the reported score while every recorded quantity stays fixed,
348 which is exactly a failure of temporal-support identifiability.

349 The result does not single out a canonical temporal-
350 support statistic, nor attribute the shift to one stage of the
351 pipeline. The predictor-side and label-side audits stay de-
352 scriptors of timing rather than estimates of recurrent sup-
353 port.

354 Where a protocol does not record the support behind a
355 score, the reported score is underdetermined. Two scores
356 at the same nominal timestamp are not performance at the
357 same effective time when the recurrent support behind them
358 differs. Comparisons across recurrent architectures should
359 therefore report or control the three latent temporal quanti-
360 ties alongside the score.

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